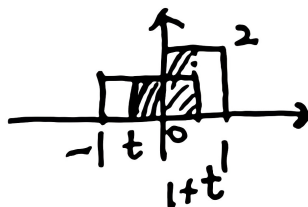


2-5 (b)

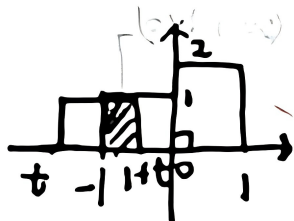
$t < -2$ 或 $t > 1$ 时 $y(t) = 0$

$-1 < t < 0$



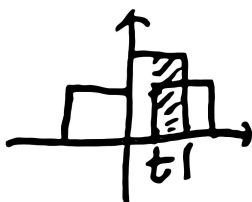
$$y(t) = -t + 2(1+t) = 2 + t$$

$-2 < t < -1$

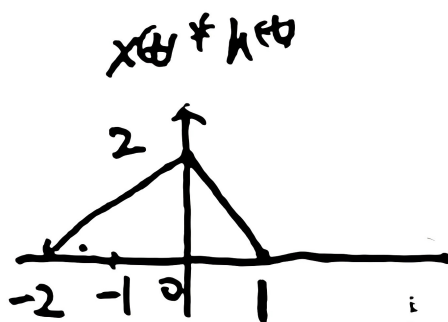


$$y(t) = t + 2$$

$0 < t < 1$



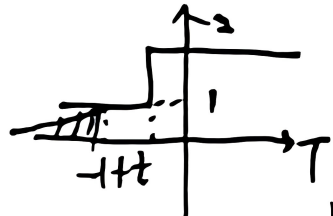
$$y(t) = 2(1-t)$$



ad1

2-5c)

$t < 0$



$$x(t) * h(t) = \int_{-\infty}^{-1+t} e^{\tau+t} d\tau$$

$= 1$

$t \geq 0$

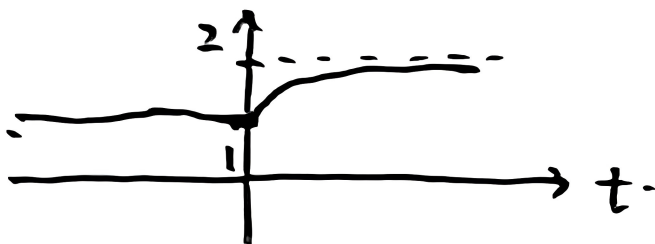


$$x(t) * h(t) = \int_{-\infty}^{-1} e^{\tau+1-t} d\tau + 2 \int_{-1}^{-1+t} e^{\tau+t} d\tau$$

$$= e^{-t} + 2(1 - e^{-t})$$

$$= 2 - e^{-t}$$

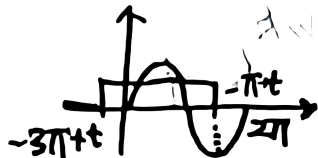
$x(t) * h(t)$



cd)

$t < \pi$ 或 $t > 5\pi$ $y(t) = 0$

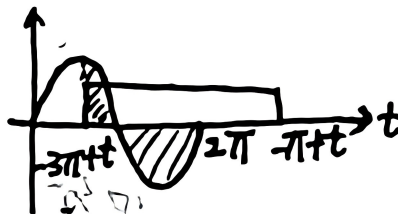
$\pi < t < 3\pi$



$$y(t) = \int_0^{-\pi+t} \sin \pi t \, dt = \cos t + 1$$

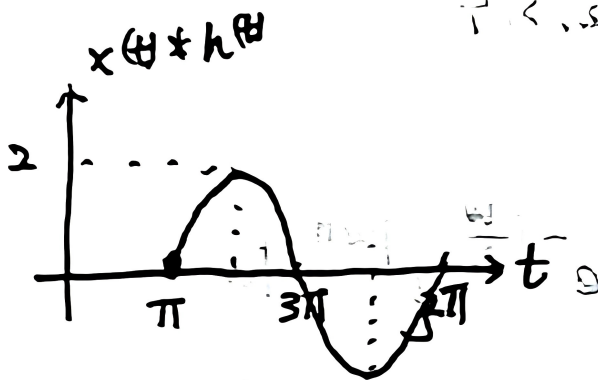
$$= \frac{\cos(t-\pi) - 1}{\pi} = \frac{1 + \cos t}{\pi}$$

$3\pi < t < 5\pi$



$$y(t) = \int_{-3\pi+t}^{2\pi} \sin \pi t \, dt$$

$$= \frac{1 - \cos(t-3\pi)}{\pi} - (\cos t + 1)$$



$$z^{-9} \quad \text{or} \quad \frac{1}{z^9} = \left(\frac{1}{z} \right)^9 = \left(z^{-1} \right)^9 = z^{-9}$$

$$x[n] * h_2[n]$$

$$= h_2[n] + \frac{1}{2} h_2[n-1] \stackrel{1-0}{=} 1-0 = 1$$

$$= \delta[n] \quad \text{for } n=0 \quad \frac{1}{2} = 0$$

$$x[n] * h_2[n] * h_1[n] = h_1[n] = \alpha^n u[n] + \beta^n u[n]$$

$$z^{-11} \quad \text{or} \quad \frac{1}{z^{11}} = \left(\frac{1}{z} \right)^{11} = \left(z^{-1} \right)^{11} = z^{-11}$$

$$h[n] = h_1[n] * (h_2[n] * h_3[n] * h_4[n]) + h_5[n]$$

(2)

$$h_3 * h_4 = h_3[n-1]$$

$$h_2[n] * h_3[n-1] = h_2[n] * h_2[n-1] = u[n]$$

$$h_1[n] * (h_2[n] * h_3[n-1])$$

$$= h_1[n] * u[n] = 4\delta[n] + 6\delta[n-1] + 7u[n-2]$$

$$= 4\delta[n] + 6\delta[n-1] + 7u[n-2]$$

$$h[n] = [8 - 4\delta[n]] u[n] - [4\delta[n-3]] u[n-3] + \delta[n] - 4\delta[n-3]$$

z-8 (1)

$$x(t) * h(t) = 3y_0(t)$$

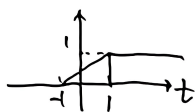


(3)

$$x(t) * h(t) = [x_0(t) * \delta(t-1)] * [h_0(t) * \delta(t-1)]$$

$$= x_0(t) * h_0(t) * \delta(t-1)$$

$$= y_0(t-1)$$



(5)

$$x(t) * h(t) = \frac{dy_0(t)}{dt}$$



$$h[n] = 4\delta[n] + 6\delta[n-1] + 7u[n-2] + \delta[n] - 4\delta[n-3]$$

$$x[n] = \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \end{matrix}$$

$$x(t) = \delta(t)$$

$$y(t) = \int_{-\infty}^t e^{-t-\tau} \delta(\tau-2) d\tau$$

$$= \begin{cases} 0 & t < 2 \\ e^{-t-2} & t \geq 2 \end{cases}$$

$$= e^{-t-2} u(t-2)$$

(2)

$$y(t) = x(t) * h(t)$$

$$= \begin{cases} 0 & t \leq 1 \\ t e^{-t+1} & 1 < t \leq 4 \\ e^{-t+4} - e^{-t+1} & t > 4 \end{cases}$$

2-13

(1)

$$\begin{aligned}
 h(t) &= h_1(t) + h_2(t) * h_3(t) * h_4(t) \\
 &= u(t) + \delta(t-1) * [\delta(t)] * u(t) \\
 &= u(t) - u(t-1)
 \end{aligned}$$

(2)

$$t < -1 \text{ 时 } x(t) * h(t) = 0$$

$$-1 \leq t < 0 \text{ 时 } x(t) * h(t) = 0$$

$$0 < t < 2 \text{ 时 } x(t) * h(t) = t+1$$

$$2 < t < 3 \text{ 时 } x(t) * h(t) = 3-t$$

$$x(t) * h(t) = \begin{cases} 0 & t < -1 \text{ 或 } t \geq 3 \\ t+1 & -1 \leq t < 0 \\ 1 & 0 \leq t < 2 \\ 3-t & 2 \leq t < 3 \end{cases}$$

(3)

$$\text{因果 } n < 0 \text{ 时 } h[n] = 0$$

$$\text{非稳定 } \sum_{n=-\infty}^{\infty} |h[n]|$$

$$\begin{aligned}
 &< \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n + \sum_{n=1}^{\infty} 1 \cdot 0.1^n \\
 &= \sum_{n=0}^{\infty} 1 \cdot 0.1^n = \infty
 \end{aligned}$$

(7)

$$\text{因果: } n < 0 \text{ 时 } h[n] = 0$$

$$\text{稳定 } \sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=-\infty}^{\infty} \frac{1}{3^n} u[n-1]$$

$$= \sum_{n=1}^{\infty} \frac{1}{3^n} = \frac{1}{3} < \infty$$

(7) X

$$h_1(t) = u(t+1) \text{ 非因果}$$

$$h_2(t) = \delta(t-2) \text{ 因果}$$

$$h_1(t) * h_2(t) = u(t-1) \text{ 因果}$$

(8) X

$$\delta(t) = \delta(t) \text{ 时 } x(t) * \delta(t) = x(t)$$

$$\text{应为 } h(t) \cdot x(t) = u(t) \text{ 时 } y(t) = u(t) * \delta(t) = u(t)$$

(9) V

$$h[n] = \delta[n] - \delta[n-1]$$

$$n < 0 \text{ 时 } h[n] = 0 \text{ 因果}$$

2-14

$$(1) \quad t < 0 \quad h(t) = 0 \text{ 因果}$$

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_2^{\infty} e^{-4t} dt < \infty$$

稳定

$$(2) \quad t < 0 \quad h(t) = 0 \text{ 不成立 非因果}$$

$$\int_{-\infty}^{\infty} |h(t)| dt = \int_{-5}^{\infty} e^{-2t} dt \text{ 稳定 (2) } \checkmark$$

$$(3) \quad t < 0 \quad h(t) = e^{6t} \neq 0 \text{ 非因果}$$

2-16

(1) V

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty \Rightarrow \lim_{t \rightarrow \infty} |h(t)| = 0$$

$$\int_{-\infty}^{\infty} |h(t)| dt = \infty$$

(3) X

$$h_1(t) = \delta(t-1) \text{ 因果}$$

$$\text{但逆系统 } h_2(t) = \delta(t+1)$$

(4) X

$$\text{如 } h[n] = 1 \quad \sum_{n=-\infty}^{\infty} |h[n]| = \infty$$

(5) V

$$\text{不为0的 } h[n] \text{ 有限}$$

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum h[n] \quad (h[n] \neq 0) < \infty$$

(6) X

$$\text{如 } h[n] = u[n] \text{ 因果但不稳定}$$

2-18

$$x_1(t) * h(t) = \frac{dy_1(t)}{dt}$$

$$\frac{dx_1(t)}{dt} * h(t) = \frac{dy_1(t)}{dt}$$

$$h(t) - h(t-1) = [u(t) - u(t-1)] - [u(t-1) - u(t-2)]$$

$$h(t) = u(t) - u(t-1)$$



$$x_1(t) * h(t) = \begin{cases} \frac{1-\cos \pi t}{\pi} & 0 < t < 2 \\ 0 & \text{其它} \end{cases}$$

其它

2-15

$$(1) \quad n < 0 \quad h[n] = 0 \text{ 因果}$$

$$\text{稳定: } \sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=0}^{\infty} \frac{1}{5^n} = \frac{5}{4} < \infty$$

$$(3) \quad \text{非因果: } n < 0 \quad h[n] = \left(\frac{1}{2}\right)^n \neq 0$$

$$\begin{aligned}
 \text{非稳定: } n < 0 \quad \sum_{n=-\infty}^{\infty} |h[n]| &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n \\
 &= \sum_{n=0}^{\infty} 2^n \\
 &= \infty
 \end{aligned}$$